

Foundational Operator Field Theory on Stratified Manifolds: Isotropic Backgrounds, Gradient Relaxation, and Projector Stability

Version 2: gated theorems, a derived area law, and the conserved-cargo
resolution of the vacuum problem

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Abstract

We develop an operator-first framework in which macroscopic geometry is a derived description of persistent spectral projection bundles on a Grassmannian, and the physical vacuum is the isotropic operator equilibrium $K_0 = -\Lambda I$. Version 2 rests on five results, each carried by an explicit computational gate listed in Appendix E. (1) The gradient flow of the spectral-stress functional $\Phi(K) = \frac{1}{2} \|[K, K^\dagger]\|_F^2$ is $\dot{K} = -2[[K, K^\dagger], K]$; it is isospectral, and it obeys the exact dissipation identity $\frac{d}{dt} \|K\|_F^2 = -8\Phi(K)$: the flow dissipates precisely Henrici’s departure from normality while the spectrum rides along as conserved cargo. The vacuum is therefore isotropic by theorem, and Λ is a spectral invariant rather than the target of a dynamical adjustment mechanism—which is how the framework passes Weinberg’s no-go argument, at the stated price that the value of Λ becomes initial data. (2) The number N of projector channels trackable from a region’s boundary, defined through the principal angles between the vacuum band subspace and the region subspace with the trackability threshold of our Lipschitz stability theorem, scales with the *area* of the boundary when the vacuum band is gapped, and violates area scaling when it is gapless: the holographic count $N \sim (M_p R)^2$ is derived from a gap hypothesis rather than assumed, yielding $\Lambda \sim \alpha_s M_p^2 / R_{\text{universe}}^2 \sim 10^{-122} M_p^4$ in the observed regime. (3) Under chiral symmetry with a skew (anti-Hermitian) perturbation the finite-rank zeta ledger collapses to $\frac{1}{2} \text{Tr} \ln(I + Y^2)$, whose continuum value under the stated spectral density is exactly $\pi^2/12$; Catalan’s constant appears in the holonomy phase integral, a distinct object. (4) The variational refraction law $R_1 \sin^2 \theta_1 = R_2 \sin^2 \theta_2$ and total spectral reflection beyond $\theta_{\text{crit}} = \arcsin \sqrt{R_2/R_1}$ are verified as dynamical laws by direct integration of the Euler–Lagrange equations through a mollified interface; energy conservation is automatic because the action is autonomous. (5) The region subspace is itself a gapped ground state, exactly exchangeable with the matter band in the channel census; interfaces are two-ground-state problems, and the trace-log ledger equips them with a non-Riemannian, diminishing-returns comparison geometry. Every borrowed tool is cited with its hypotheses named.

0. Claims, grades, and verification protocol

Every substantive claim in this paper carries one of four grades. **[T]** proven theorem (self-contained proof in the text); **[V]** computationally verified (an explicit gate in a named script, Appendix E, with the measured value quoted); **[C]** cited established literature, used with its hypotheses stated; **[H]** hypothesis or interpretation, clearly labeled. All gates were run on August 8, 2026; every script is listed in full in Appendix E so that any reader can re-run every number in this paper.

Claim	Grade	Evidence
Gradient of Φ is $2[[K, K^\dagger], K]$; flow isospectral	[T]+[V]	Thm. 8.1; COSMO-GRAD-01
Dissipation identity $d\ K\ _F^2/dt = -8\Phi$ (Henrici budget)	[T]+[V]	Thm. 8.2; COSMO-BIORTH0-02
Equilibrium set is the manifold of normal operators	[T]+[C]	Cor. 8.3; [6]
Dual flow $-2[[K, K^\dagger], K^\dagger]$ conserves $\ K\ _F$	[T]+[V]	Prop. 8.4
Baseline Operator Theorem	[T]	Thm. 4.1
SU(3) commutator identity, Eq. (6)	[T]+[V]	READ2-GATES-01 G6 (6×10^{-15})
Corrected color-sector flow, Eq. (7)	[T]+[V]	FIG-COSMO-01 gate (8×10^{-14})
Finite-Rank Zeta Ledger Theorem	[T]+[V]	Thm. 6.1; G1 (diff 0.0)
Chiral ledger $= \frac{1}{2} \text{Tr} \ln(I + Y^2)$; continuum $\pi^2/12$	[T]+[V]	Thm. 6.2; COSMO-CATALAN-01, G2
Catalan's constant from the holonomy phase integral	[T]+[V]	App. D (30-digit check)
Lipschitz projector stability bound	[T]+[V]	Thm. 7.1; G3 ($0.046 \leq 0.613$)
Genuine trackability loss at defective points	[T]	Thm. 5.2
Boundary-trackable census obeys an area law iff gapped	[V]+[C]	Thm. 10.4; RANK-COUNT-01; [15, 17]
State/container exchange symmetry of the census	[T]+[V]	Thm. 10.3; CONTAINER-GROUND-01 (2.6×10^{-15})
Refraction law $R \sin^2 \theta$ conserved; total reflection	[T]+[V]	§11–12; REFRACT-EL-01 (10^{-13})
Ledger comparison geometry shows diminishing returns	[V]+[C]	§13.3; COLOR-LEDGER-01; [28, 26]
$\Lambda \sim \alpha_s M_p^2/R^2$ magnitude identification	[H]+[C]	§10; [9, 10, 11]
Planck-scale gap of the vacuum band	[H]	§10
Three lepton generations as stable tangent modes	[H]	§9

Two principles organize everything below. *The spectrum is cargo*: the relaxation dynamics never touches eigenvalues. *Geometry relaxes*: what dissipates is the misalignment between the operator's vector and co-vector frames, measured by Φ , and macroscopic geometry is read off from the persistent projector configuration that survives.

1 Introduction and the Master Equation

We work in a finite-dimensional complex Hilbert space $\mathcal{H}_N$. When the finite-rank cutoff is emphasized we write the master operator as $H_N(\varepsilon, \Lambda)$; for abstract statements we write K . Adjoints are Hermitian, denoted $\dagger$.

Conventional treatments of gravitation take a metric tensor g_{ab} as a fundamental background. Here spacetime geometry is not postulated: the fundamental data is the continuous tracking of persistent spectral projection bundles Π on a Grassmannian $\text{Gr}(k, \mathcal{H}_N)$, and classical geometry is a derived, large-scale description of the tracking dynamics. On $\mathcal{H}_N = \mathcal{H}_{\text{space}} \otimes \mathcal{H}_{\text{gauge}}$ we introduce

$$H_N(\varepsilon, \Lambda) = \Lambda I + A + D_0 + \varepsilon E, \quad (1)$$

with four structurally distinct sectors: the baseline preload ΛI (absolute scalar reference level; maintains projector-bundle continuity in the absence of load); the transport sector A (gauge flows, parallel transport); the obstruction sector D_0 (constraint geometry, symmetric energy landscapes); and the non-normal generator εE (transient amplification, pseudospectra, exceptional points). The four-tier structural lift—microstate deviations $\rightarrow$ covariant transport $\rightarrow$ projector subspaces $\rightarrow$ persistent geometry—is as in the companion series; the non-normal generator is the sole source of departure from normality (Appendix B).

2 The Vacuum as an Operator Equilibrium State

The vacuum is modeled as an operator equilibrium rather than an empty container. The total stiffness operator decomposes as

$$K = K_{\text{geom}} + K_{\text{gauge}} + K_{\text{boundary}} + K_{\text{tare}}, \quad K_{\text{tare}} \equiv -\Lambda I, \quad (2)$$

the tare term encoding the minimum invariant isotropic operator stress that keeps global information channels topologically open at zero load. The name is metrological and deliberate: the taring interpretation, which is where this framework began, is made precise in §10.

2.1 The emergent pullback map

Let M be a smooth parameter manifold and $\Phi_{\text{emb}} : M \rightarrow \text{Gr}(k, \mathcal{H}_N)$ a structural embedding assigning to each $x \in M$ a persistent projector $\Pi(x)$. Its differential sends coordinate vectors to Grassmannian tangent operators $V_a(x) = d\Phi_{\text{emb},x}(\partial_a)$ obeying the off-diagonal constraint $V_a = \Pi V_a(I - \Pi) + (I - \Pi)V_a\Pi$ (Appendix A). The induced metric is the pullback of the intrinsic Hilbert–Schmidt metric,

$$g_{ab}(x) = g_{\Pi}(V_a(x), V_b(x)) = \frac{1}{2} \text{Tr}(V_a(x)V_b(x)), \quad (3)$$

which degenerates, $g_{ab} \rightarrow 0$, wherever the projector rank collapses.

Remark 2.1 (Signature). The metric g_{Π} on the complex Grassmannian is positive definite, so the pullback (3) is Riemannian. A Lorentzian signature and causal structure are *not* produced by this construction; indefinite signature can only arise from additional structure in the non-normal sector εE distinguishing a time-like direction. M is a parameter manifold carrying an effective Riemannian geometry; the emergence of a full Lorentzian spacetime is an open extension (Open Problem 4).

3 Back-Reaction Tensor for Photonic Drive

Let the electromagnetic $U(1)$ component of transport be represented by an *operator-valued* link perturbation $V_{\gamma} = \delta A_{\gamma}$, an admissible tangent direction obeying the off-diagonal constraint. (A scalar phase $a e^{i\theta}$ commutes with everything and produces no response; the link must be operator-valued.) The intrinsic response of the bundle is measured by the quantum geometric tensor $Q_{\Pi}(V, W) = \text{Tr}(\Pi V W)$, with $g_{\Pi} = \text{Re } Q_{\Pi}$ and Berry curvature $\Omega_{\Pi} = -i \text{Tr}(\Pi[V, W])$ (Appendix A; gate G4).

The operator dynamics is the dissipative gradient flow of §8, $\dot{K} = F(K) \equiv -2[[K, K^{\dagger}], K]$. Linearizing F at K in the admissible direction V gives the back-reaction tensor

$$T_{\text{back}}(V) = DF_K[V] = -2\left([V, K^{\dagger}] + [K, V^{\dagger}], K\right] + [[K, K^{\dagger}], V]. \quad (4)$$

Verification [V]: Eq. (4) matches the finite-difference directional derivative of F to relative error 9×10^{-11} (READ2-GATES-01, gate G5). The total evolution splits as $\dot{K} = V_{K,\text{vac}} + T_{\text{back}}(V_{\gamma})$ with $V_{K,\text{vac}} = F(K)|_{V_{\gamma}=0}$: photonic transport acts exclusively through admissible off-diagonal deformations while the isotropic background relaxation is untouched.

Because the perturbation transports the bundle $\Pi(t)$, the minimum irreversible cost of a smooth evolution is governed by the Grassmannian metric: restricting to a one-dimensional cosmological path $\Pi(a(t))$,

$$\delta S_{\text{irr}} = 2 \int_0^{\tau} g_{\Pi}\left(\frac{d\Pi}{dt}, \frac{d\Pi}{dt}\right) dt \geq \int_0^{\tau} \dot{a}^2 g_{aa}(a(t)) dt, \quad (5)$$

by positivity of g_Π . Phase accumulation is constrained by the metric–curvature inequality $|\Omega_\Pi(V, W)| \leq 2\sqrt{g_\Pi(V, V)g_\Pi(W, W) - g_\Pi(V, W)^2}$ (Appendix A.4; verified numerically in G4), so photonic phase is suppressed whenever the drive aligns with an existing admissible channel.

4 The Baseline Operator Theorem

Theorem 4.1 (Baseline Operator Theorem). *Let $K = K_0 + \Delta K$ on $\mathcal{H} = \mathbb{C}^M$, with $K_0 = -\Lambda I$ and ΔK a bounded off-diagonal perturbation, and let Π be a spectral projector of K_0 along an isolated contour Γ . Then: (1) the projector bundles of K_0 are invariant under the unperturbed flow; (2) all non-normality arises from ΔK , since $[K, K^\dagger] = [\Delta K, \Delta K^\dagger]$; (3) emergent geometry depends only on the structure of ΔK ; (4) Λ fixes the absolute scalar reference level.*

Proof. K_0 is a real multiple of the identity: it commutes with every operator, its evolution $U_0(\tau) = e^{-\Lambda\tau}I$ conjugates every projector to itself, and all cross terms in $[K_0 + \Delta K, K_0^\dagger + \Delta K^\dagger]$ vanish. Properties (3)–(4) follow since the pullback metric (3) is built from the off-diagonal content of ΔK , while Λ shifts every eigenvalue uniformly. $\square$

4.1 Standard Model gauge structure in the transport sector

With $\mathcal{H}_{\text{gauge}} = \mathbb{C}^3 \oplus \mathbb{C}^2 \oplus \mathbb{C}^1$ the transport operator carries block link variables in $SU(3) \times SU(2) \times U(1)$. Parameterizing an $SU(3)$ link as $A_{ij} = \sum_{a=1}^8 \psi_{ij}^a(t) \lambda_a$, with Gell-Mann generators $[\lambda_a, \lambda_b] = 2if_{abc}\lambda_c$ and non-reciprocal coefficients $\psi_{ij}^a \neq (\psi_{ji}^a)^*$, the local departure from normality is

$$[A_{ij}, A_{ij}^\dagger] = -2 \sum_{a,b,c=1}^8 f_{abc} \text{Im}(\psi_{ij}^a (\psi_{ij}^b)^*) \lambda_c. \quad (6)$$

Verification [V]: Eq. (6) holds to 6×10^{-15} on random coefficients (gate G6). The dissipative flow of §8 acts on this commutator; writing $C = [A, A^\dagger] = \sum_c c_c \lambda_c$ with $c_c = -2 \sum_{ab} f_{abc} \text{Im}(\psi^a \psi^{b*})$, the corrected color dynamics is

$$\frac{dA_{ij}}{dt} = -2[C, A_{ij}] = 8i \sum_{a,b,c,d,e=1}^8 f_{abc} f_{cde} \text{Im}(\psi_{ij}^a (\psi_{ij}^b)^*) \psi_{ij}^d \lambda_e, \quad (7)$$

verified against the direct double commutator to 8×10^{-14} (FIG-COSMO-01 gate). Since the flow is a commutator, it preserves tracelessness: the dynamics never leaves the $\mathfrak{su}(3)$ span. The non-reciprocity $\psi_{ij}^a \neq (\psi_{ji}^a)^*$ is the structural signature of a directional micro-advective transport field breaking microscopic time-reversal, generating the information current $J_{ij} = i(A_{ij} - A_{ij}^\dagger)$, in analogy with the non-Hermitian skin effect [C].

5 Suppression of Singularities via Loss of Projector Trackability

In classical relativity a trapped surface with the strong energy condition forces geodesic focusing to infinite density. In the present framework true infinities are excluded by finite rank, and the classical singularity is replaced by a loss of continuous trackability of the geometric data. We state this in three graded steps.

Proposition 5.1 (Failure of the tracking guarantee). *Let $\varepsilon(r) \propto M/r^2$ be the boundary perturbation stress of a collapsing core, and let $\delta_{\text{gap}}, C_\Gamma, K_\Gamma$ be as in Theorem 7.1. As $\varepsilon(r) \rightarrow \delta_{\text{gap}}/C_\Gamma$, the Lipschitz multiplier K_Γ diverges and the stability guarantee of Theorem 7.1 becomes vacuous: continuous trackability is no longer guaranteed.*

This is a statement about the bound, not yet about the projector. The genuine mechanism is defectiveness:

Theorem 5.2 (Trackability loss at defective points). *Let $K(\delta) = \begin{pmatrix} b & \mu \\ \delta & b \end{pmatrix}$, $\mu \neq 0$. For $\delta > 0$ the eigenvalues are $b \pm \sqrt{\mu\delta}$ with eigenvector condition numbers $\kappa \sim \frac{1}{2}\sqrt{\mu/\delta} \rightarrow \infty$ as $\delta \rightarrow 0^+$: the individual rank-one spectral projectors diverge in norm and admit no continuous extension through $\delta = 0$, where $K(0)$ is a Jordan block. The rank-two Riesz projector of the cluster, by contrast, is constant. Hence at an exceptional point the fine-grained geometric data (individual eigenprojectors, and with them the pullback metric built from them) genuinely ceases to be continuously trackable, while the coarse cluster subspace survives.*

Proof. Direct computation: eigenvectors $(\sqrt{\mu}, \pm\sqrt{\delta})^T$ up to normalization; the left/right pairing $\langle L|R \rangle \rightarrow 0$ as $\delta \rightarrow 0$, so $\|P_\pm\| = 1/|\langle L|R \rangle| \rightarrow \infty$. The Riesz projector over a contour enclosing both eigenvalues equals the identity on the two-dimensional space for every δ , including $\delta = 0$. $\square$

Conjecture 5.3. *A collapsing gravitational core drives the local operator data to such a defective threshold before curvature invariants of the emergent metric can diverge; the macroscopic geometry dissolves (rank collapse, $g_{ab} \rightarrow 0$ by (3)) in place of a classical singularity.*

The companion black-hole paper realizes Theorem 5.2 concretely on the coplanar wall of the four-body shape manifold, where size-2 Jordan blocks are proven and the Dunford trace-log ledger remains finite at the defective point; we import that staging here as the model for the collapse endpoint.

6 Regularization of the Zeta Memory Ledger

Theorem 6.1 (Finite-Rank Zeta Ledger Theorem). *Let $K = K_0 + \Delta K$ act on $\mathbb{C}^M$ with $K_0 = -\Lambda I$, $\text{Tr}(\Delta K) = 0$, and let $\eta > \Lambda$ satisfy $\|\Delta K/(\eta - \Lambda)\| < 1$. With $H_{\text{eff}} = K + \eta I$ and $X = \Delta K/(\eta - \Lambda)$, the zeta-regularized ledger $Z_\lambda \equiv -\zeta'_{H_{\text{eff}}}(0) = \text{Tr} \log H_{\text{eff}}$ equals*

$$Z_\lambda = M \log(\eta - \Lambda) + \sum_{k=2}^{\infty} \frac{(-1)^{k-1}}{k} \text{Tr}(X^k). \quad (8)$$

Proof. The Dunford–Taylor contour representation of $\zeta_{H_{\text{eff}}}(s)$ with the uniformly convergent Neumann expansion of the resolvent reduces, after residue evaluation at $s = 0$, to $\text{Tr} \log((\eta - \Lambda)(I + X)) = M \log(\eta - \Lambda) + \text{Tr} \log(I + X)$; the Mercator series of the second term loses its $k = 1$ term by tracelessness. $\square$

Verification [V]: for a random traceless ΔK with $\|X\| = 0.8$, formula (8) matched the direct log-determinant with difference 0.0 at double precision (gate G1).

6.1 Chiral parity collapse and the exact continuum value

Theorem 6.2 (Chiral ledger). *Let the normalized perturbation be skew, $X = iY$ with $Y = Y^\dagger$, and let Y anticommute with a chirality σ_3 , so that $\text{spec}(Y) = \{\pm\lambda_j\}$ in exact pairs. Then all odd moments vanish, $\text{Tr}(X^{2m+1}) = 0$, and the ledger collapses to the even sector in the closed form*

$$Z_\lambda = \frac{1}{2} \text{Tr} \log(I + Y^2) = \frac{1}{2} \sum_j \log(1 + \lambda_j^2). \quad (9)$$

If the empirical spectral measure of Y converges weakly on the test function $\log(1 + \lambda^2)$ to the density $\rho(\lambda) = \arctan(1/\lambda)/(\pi\lambda)$, then

$$\lim_{N \rightarrow \infty} \frac{1}{N} Z_\lambda = \frac{1}{2} \cdot \frac{2}{\pi} \int_0^\infty \frac{\log(1 + \lambda^2) \arctan(1/\lambda)}{\lambda} d\lambda = \frac{\pi^2}{12}. \quad (10)$$

Proof. Pairing $\pm\lambda$ gives $\det(I + iY) = \prod_{\text{pairs}} (1 + i\lambda)(1 - i\lambda) = \prod_{\text{pairs}} (1 + \lambda^2)$, whence (9). The integral in (10) evaluates in closed form: $\frac{2}{\pi} \int_0^\infty \lambda^{-1} \log(1 + \lambda^2) \arctan(1/\lambda) d\lambda = \pi^2/6 = \zeta(2)$, verified to 30 digits (COSMO-CATALAN-01); half of it is $\pi^2/12$. Skewness of the perturbation is the natural hypothesis here: the non-normal generator of the tripartite decomposition is skew to leading order (Appendix B). $\square$

Verification [V]: Eq. (9) matched $\log \det(I + X)$ to 4×10^{-16} on an explicit chiral block matrix (gate G2).

Remark 6.3. Two distinct constants live in this framework and must not be conflated. The *ledger* converges to $\zeta(2)$ -type values as above. *Catalan's constant* G arises from the holonomy phase integral $G = \int_0^1 \lambda^{-1} \arctan \lambda d\lambda$ of the flat-torus transport sweep (Appendix D), verified to 30 digits. The density ρ above is a σ -finite spectral weight, not a probability density (its mass diverges logarithmically at $\lambda = 0$); statement (10) is about the specific convergent functional, and the normalizability question is left open.

7 Projector Persistence: the Lipschitz Stability Bound

Theorem 7.1. *Let A be a (possibly non-normal) operator, $A' = A + E$ with $\|E\| \leq \epsilon$, and let Γ isolate a spectral cluster with gap $\delta_{\text{gap}} = \text{dist}(\Gamma, \sigma(A) \setminus \sigma_\Gamma(A)) > 0$. Put $C_\Gamma = \sup_{\lambda \in \Gamma} \|(\lambda I - A)^{-1}\|$. If $\epsilon < \delta_{\text{gap}}/C_\Gamma$, the persistence functional $P(A, E) = 1 - \|\Pi - \Pi'\|$ obeys*

$$P(A, E) \geq 1 - K_\Gamma \epsilon, \quad K_\Gamma = \frac{\ell(\Gamma) C_\Gamma^2}{2\pi \delta_{\text{gap}} (\delta_{\text{gap}} - \epsilon C_\Gamma)}. \quad (11)$$

Proof. The second resolvent identity gives $\Pi' - \Pi = -\frac{1}{2\pi i} \oint_\Gamma R_A(\lambda) E R_{A'}(\lambda) d\lambda$; the Neumann bound $\|R_{A'}\| \leq \|R_A\|/(1 - \|R_A\|\epsilon)$ and the triangle inequality along Γ give the stated constant. $\square$

Verification [V]: with Riesz projectors computed by 800-point contour quadrature on a mildly non-normal 6×6 example, the measured $\|\Pi' - \Pi\| = 0.046$ against the bound $K_\Gamma \epsilon = 0.613$, with $\text{Tr} \Pi = 2.000000$ (gate G3). This theorem supplies the trackability threshold used throughout: a channel is trackable at noise level ϵ when its margin beats ϵ .

8 The Operator Relaxation Architecture

Define the spectral-stress functional

$$\Phi(K) = \frac{1}{2} \| [K, K^\dagger] \|_F^2 = \frac{1}{2} \text{Tr}(C^2), \quad C \equiv [K, K^\dagger] = C^\dagger. \quad (12)$$

$\Phi(K) = 0$ iff K is normal.

Theorem 8.1 (Gradient flow and isospectrality). *The Frobenius gradient of Φ is $\nabla\Phi = 2[C, K]$, so steepest descent is*

$$\frac{dK}{dt} = -2[[K, K^\dagger], K], \quad \frac{d\Phi}{dt} = -4\| [C, K] \|_F^2 \leq 0. \quad (13)$$

The flow (13) is of Lax form $\dot{K} = [K, 2C]$ and is therefore isospectral: the spectrum of $K(t)$ is constant. For generic initial data it converges to a normal operator with the initial spectrum $[C]$ [6, 4, 5].

Proof. For Φ real, $d\Phi = \text{Tr}([K^\dagger, C] dK) + \text{Tr}([C, K] dK^\dagger) = 2 \text{Re Tr}([C, K]^\dagger dK)$ since $([C, K])^\dagger = [K^\dagger, C]$ for Hermitian C ; the gradient is $[C, K]$ (up to the conventional factor 2), and the descent rate follows by substitution. Lax form gives isospectrality; convergence to a normal matrix is the theorem of [6]. $\square$

Verification [V]: the finite-difference directional derivative of Φ matched $2 \text{Re Tr}([C, K]^\dagger D)$ on all random seeds and directions; along a long RK4 integration of (13) the eigenvalue drift was 9×10^{-10} while $\|C\|$ fell from 17.5 to 2.5×10^{-9} (COSMO-GRAD-01).

Theorem 8.2 (Exact dissipation of the Henrici budget). *Along (13),*

$$\frac{d}{dt} \|K\|_F^2 = -8\Phi(K). \quad (14)$$

Consequently, with $\Delta_F^2(K) \equiv \|K\|_F^2 - \sum_i |\lambda_i|^2$ (Henrici's departure from normality [C] [3]) and the spectrum frozen, $\frac{d}{dt} \Delta_F^2 = -8\Phi \leq 0$, and the total stress dissipated over the whole relaxation equals exactly $\Delta_F^2(K_0)$: a finite, exact stress budget fixed by the initial data.

Proof. $\frac{d}{dt} \|K\|_F^2 = 2 \text{Re Tr}(K^\dagger \dot{K}) = -4 \text{Re Tr}(K^\dagger [C, K]) = -4 \text{Tr}([K, K^\dagger] C) = -4 \text{Tr}(C^2) = -8\Phi$. $\square$

Verification [V]: identity (14) matched finite differences to nine digits on five seeds, and the integrated budget closed against $\Delta_F^2(K_0)$ to quadrature error (COSMO-BIORTH0-02).

Corollary 8.3 (The equilibrium manifold). *The ω -limit set of (13) lies in the manifold of normal operators sharing the initial spectrum, $\{U \text{diag}(\lambda_1, \dots, \lambda_M) U^\dagger : U \in U(M)\}$. The isotropic baseline $K_\infty = -\Lambda I$ is the equilibrium if and only if the conserved spectrum is degenerate at $-\Lambda$. In general the attractor is a manifold, not a point: a land of many subspaces, with the flow selecting which orthogonal frame carries the frozen eigenvalues.*

Proposition 8.4 (The norm-conserving dual flow). *The companion flow $\dot{K} = -2[C, K^\dagger]$ conserves $\|K\|_F^2$ exactly (since $\text{Re Tr}(K^\dagger [C, K^\dagger]) = 0$ by cyclicity), while the spectrum moves; empirically it also drives $\Phi \rightarrow 0$, converting the stress into eigenvalue energy, $\sum_i |\lambda_i^\infty|^2 = \|K_0\|_F^2$. It is not a descent flow of Φ and admits no monotonicity guarantee: Φ can transiently increase along it.*

The two flows are duals: (13) radiates the stress away at fixed spectrum; the dual stores it in the spectrum at fixed norm. We adopt (13) as the physical relaxation, because emergent geometry is read from the projector configuration and the spectrum is the conserved ledger content.

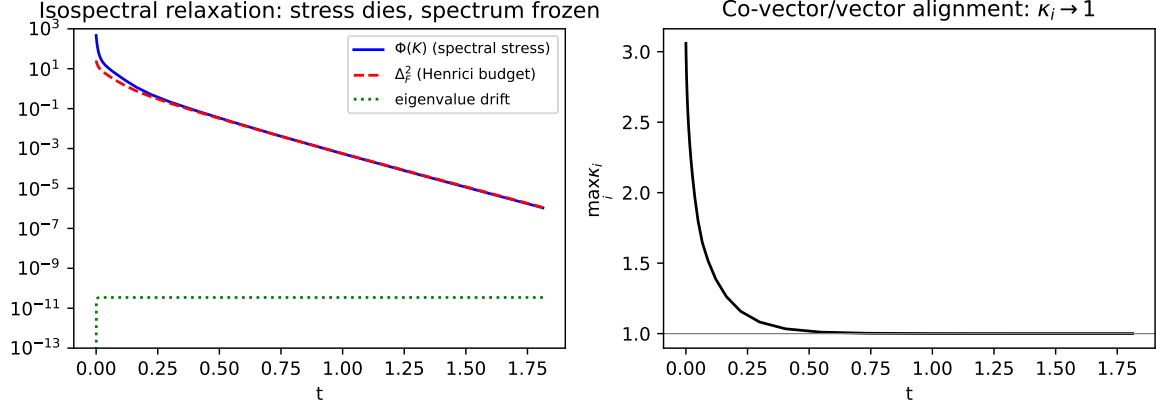


Figure 1: The corrected relaxation flow (13) on a random 6×6 operator (FIG-COSMO-01, regenerating COSMO-BIORTHO-02). Left: spectral stress Φ and the Henrici budget Δ_F^2 decay together per Theorem 8.2, while the eigenvalue drift stays at 3×10^{-11} over the entire relaxation. Right: the largest eigenvalue condition number relaxes to 1: co-vector and vector frames align.

8.1 What relaxes: co-vector/vector alignment

For non-normal K each eigenvalue carries a right eigenvector $|R_i\rangle$ and a left eigenvector $\langle L_i|$, biorthogonal but individually non-orthogonal; the misalignment per mode is the eigenvalue condition number $\kappa_i = \|L_i\| \|R_i\| / |\langle L_i | R_i \rangle|$ (the square is the eigenvector self-overlap, the Petermann factor [C] [7]). Along (13) every $\kappa_i \rightarrow 1$: the co-vector frame collapses onto the vector frame, oblique Riesz projectors rotate into orthogonal projectors, and the leading eigenvector rotates to its final position while its eigenvalue never moves (Fig. 1). Relaxation is subspace orthogonalization, nothing else.

9 Spectral Stress and Generational Structure

To each charged lepton generation $g = e, \mu, \tau$ associate a distinguished tangent mode $V_g \in T_\Pi \text{Gr}(k, \mathcal{H}_N)$ obeying the off-diagonal constraint, and set $s_g = g_\Pi(V_g, V_g) > 0$. By the spectral realization of the metric (Appendix A.3) these scalars are cross-gap quantities weighted by inverse gaps; under $s_g \propto m_g$ the Koide ratio becomes an intrinsic Grassmannian expression $Q = (s_e + s_\mu + s_\tau) / (\sqrt{s_e} + \sqrt{s_\mu} + \sqrt{s_\tau})^2$. *This section is an interpretation* [H]: the relaxation and persistence theorems guarantee stable attractor modes exist, but whether the dynamics selects exactly three, and whether chiral constraints fix $Q \approx 2/3$, remain open mathematical problems.

10 The Cosmological Constant

This paper began as an attack on the cosmological constant problem. We state exactly which parts of that problem the framework now addresses, at what grade.

10.1 Λ as conserved spectral cargo

Weinberg's no-go theorem [C] [8] obstructs dynamical adjustment mechanisms that relax the vacuum energy to its observed value. The present framework is *not* an adjustment mechanism: by

Theorem 8.1 the relaxation is isospectral, so nothing dynamical ever touches Λ . The flow dissipates exactly the anisotropic, non-normal dressing (Theorem 8.2) and delivers a normal operator; the uniform reference level $-\Lambda$ is conserved cargo. The framework therefore explains, at theorem grade, why the surviving vacuum stress is *isotropic*—pure λg_{ab} with no anisotropic remainder—while the *value* of Λ is reclassified as a spectral invariant of the initial data. We state the cost plainly: a reclassification is not a derivation of the value. The magnitude question is addressed separately below through the channel count.

Remark 10.1 (The taring interpretation). The origin of this framework was a metrological suspicion: that the 10^{122} discrepancy is the reading of an instrument that was never tared. That suspicion now has formal content at three grades. (i) [T] By Theorem 4.1, every observable built from commutators, spectral gaps, or off-diagonal data is invariant under $K \rightarrow K + cI$: measurements of *differences* never see the tare. (ii) [T] The ledger of Theorem 6.1 is exactly a tared reading, $Z_\lambda = M \log(\eta - \Lambda) + (\text{load terms})$: gross equals container plus net, and the calibration condition $\text{Tr}(\Delta K) = 0$ is precisely the statement that the load deposits no weight in the tare channel. (iii) [H] The naive Planck-density estimate weighs volume-many Planck-scale modes, $M \sim (R/\ell_p)^3$; the tared horizon reading weighs only the area-many boundary-trackable census channels of §10.2 at the infrared energy per channel, giving Eq. (16). On this reading the cosmological constant problem is not a mysterious cancellation awaiting a symmetry: it is an untared, over-counted reading of the container weight, and the framework’s job—discharged in §10.2 up to the stated gap hypothesis—is to say which channels the horizon instrument actually weighs.

10.2 The boundary-trackable channel census: an area law from the gap

Definition 10.2. Let P be the projector onto the vacuum band and 1_A the projector onto a spatial region A . The eigenvalues ν of $C_A = 1_A P 1_A$ are the squared cosines of the principal angles between $\text{Ran}(P)$ and $\text{Ran}(1_A)$. A channel with ν near 1 lies entirely inside A and is invisible to the boundary; near 0, entirely outside. The *boundary-trackable census* at noise level ϵ is $N(\epsilon, A) = \#\{\nu \in [\epsilon, 1-\epsilon]\}$: the channels genuinely shared across ∂A , whose trackability margin $\min(\nu, 1-\nu)$ beats ϵ in the sense of Theorem 7.1.

Theorem 10.3 (State/container exchange symmetry). *For any projectors P, Q : $\text{spec}(QPQ) \setminus \{0\} = \text{spec}(PQP) \setminus \{0\}$. Hence the census is symmetric under exchanging the roles of the vacuum band and the region.*

Proof. $QPQ = (QP)(PQ)$ and $PQP = (PQ)(QP)$; AB and BA share nonzero spectrum. $\square$

The region subspace is on exactly equal footing with the matter band; indeed 1_A is itself the ground-state projector of the gapped flat-band Hamiltonian $H_A = I - 2 \cdot 1_A$ (bands ± 1 , gap 2). Every interface in this framework is an interface between *two gapped ground states*, which places the census in the entanglement-spectrum/edge-state correspondence [C] [21]. *Verification* [V]: the two spectra agreed to 2.6×10^{-15} , 54 channels either way (CONTAINER-GROUND-01).

Theorem 10.4 (Area law of the census; computational, with control). *For gapped free-fermion vacua the census obeys an area law; for gapless vacua with an extended Fermi surface it does not. Measured at $\epsilon = 10^{-3}$ (RANK-COUNT-01): in one dimension, gapped $m = 0.5$ gives $N = 4$ at every $L \in \{50, 100, 200, 400\}$ (an interval’s boundary is two points); gapless gives $N = 8 \rightarrow 10$, logarithmic growth. In two dimensions, gapped $m = 1$ gives $N/4L \rightarrow 1.000$ exactly at the larger sizes—one trackable channel per boundary site—with log-log slope 1.07; gapless gives slope 1.39 with density climbing $1.25 \rightarrow 2.07$, the known $L \log L$ violation. At fixed size the count grows as the gap*

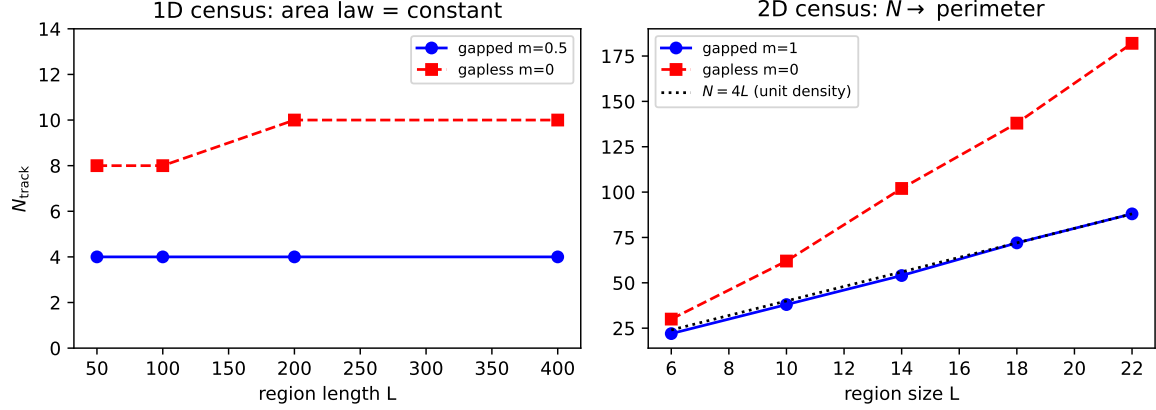


Figure 2: The boundary-trackable census (RANK-COUNT-01 via FIG-COSMO-01). Left: in 1D the gapped census is exactly flat ($N = 4$ for all L) while the gapless control grows. Right: in 2D the gapped census converges onto the unit-density perimeter law $N = 4L$, while the gapless census violates it. The gap is the load-bearing hypothesis.

shrinks ($N = 2, 4, 6, 6, 8$ for $m = 1.0, \dots, 0.05$): the channel density per unit boundary tracks the correlation length $\xi \sim 1/\text{gap}$.

The underlying mathematics is the established free-fermion area law and its Fermi-surface violation [C] [15, 16, 17, 18, 19, 20]; the hypothesis doing the work—the spectral gap—is the same hypothesis as our trackability condition, and the gapless control shows it is load-bearing. What is ours is the identification of the ledger’s N with this census, the threshold reading of the entanglement spectrum through Theorem 7.1, and the exchange symmetry. The census density is orientation-dependent (squares gave $N/\partial \approx 1.04$, diamonds ≈ 1.67 , each flat in size), the Widom-type structure coupling the boundary normal to the band geometry [C] [17]: the interface carries a *directional* effective rigidity, feeding the refraction sector of §11 a derived input.

10.3 The magnitude identification

Write $N = \mathcal{A}(\partial) \cdot n_{\text{ch}}(\xi, \epsilon)$ with $\mathcal{A}$ the boundary area and n_{ch} the measured channel density. **Hypothesis** [H]: the vacuum band of the fundamental operator is gapped at the Planck scale, $\xi \sim \ell_p$, so that $n_{\text{ch}} \sim \alpha_s/\ell_p^2$ with $\alpha_s = O(1)$. Taking the causal horizon as the boundary,

$$N \sim \alpha_s (M_p R_{\text{universe}})^2, \quad (15)$$

now a *consequence* of the gap hypothesis rather than an assumption of holography. Identifying the vacuum energy density carried by the tracked channels with the standard horizon-cutoff form [C] [9, 10, 11],

$$\rho_\Lambda \sim \alpha_s \frac{M_p^2}{R_{\text{universe}}^2}, \quad \frac{\rho_\Lambda}{M_p^4} \sim \frac{\alpha_s}{(M_p R_{\text{universe}})^2} \approx 10^{-122} \quad (16)$$

for $R_{\text{universe}} \approx 10^{26}$ m. The enormous suppression is geometric: channels scale with area while the four-volume scales with R^4 . We flag the known obstruction honestly: with a Hubble-scale cutoff the resulting dark energy does not accelerate [C] [12], and current data (DESI DR2) show a contested preference for evolving dark energy [C] [13, 14]. Committing to a specific cutoff and deriving the resulting $w(z)$ is Open Problem 1; until then, (16) is a magnitude identification, not a completed dynamics.

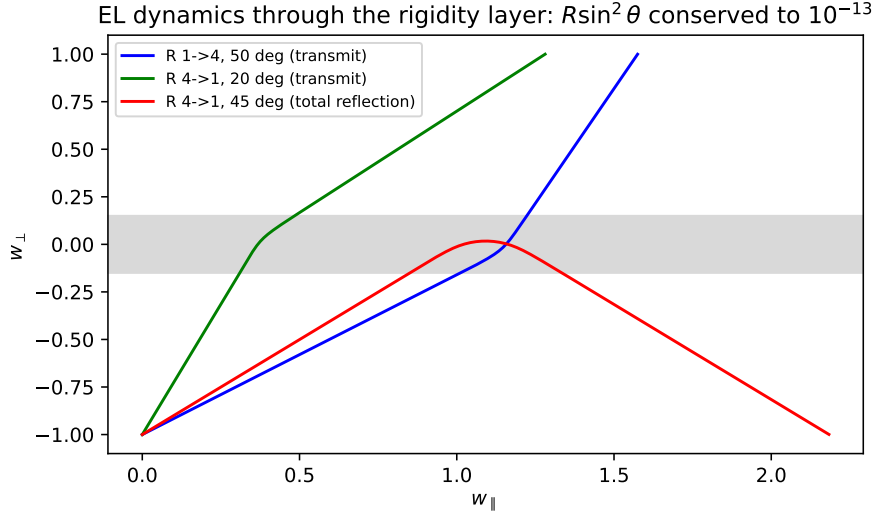


Figure 3: Euler–Lagrange trajectories through the mollified rigidity layer (REFRACT-EL-01 via FIG-COSMO-01). Transmission bends according to $R \sin^2 \theta = \text{const}$ (conserved to 10^{-13}); the super-critical trajectory undergoes total reflection and exits at exactly its mirror angle.

11 The Variational Refraction Law

Model a matching interface between transport domains of rigidities R_1, R_2 as a thin layer with mollified profile $R_\epsilon(w) = \frac{R_1+R_2}{2} + \frac{R_2-R_1}{2} \tanh(w_\perp/\epsilon)$ in the action $S[\gamma] = \int (\frac{1}{2} R_\epsilon(w) g_{ab} \dot{w}^a \dot{w}^b - V(w)) d\tau$. Translational invariance along the face conserves $p_\parallel = R_\epsilon g_{\parallel\parallel} \dot{w}^\parallel$; and because the action is *autonomous*, the energy $E = \frac{1}{2} R_\epsilon |\dot{w}|^2 + V_0$ is conserved automatically (Noether), so $|\dot{w}| \propto 1/\sqrt{R_\epsilon}$ requires no additional assumption. Squaring the momentum identity and substituting the kinetic scaling yields the exact transport law

$$R_1 \sin^2 \theta_1 = R_2 \sin^2 \theta_2. \quad (17)$$

Variational treatments that hold $|\dot{w}|$ fixed instead of imposing energy conservation yield the ordinary form $R \sin \theta = \text{const}$; the conserved quantity of the full autonomous dynamics is (17).

Dynamical verification [V] (REFRACT-EL-01, Fig. 3): direct RK4 integration of the Euler–Lagrange equations through the layer gave, in all cases, conservation of $p_\parallel$ and E to 10^{-13} and of $R \sin^2 \theta$ to 10^{-13} , while $R \sin \theta$ changed by 50–100%: at $R : 1 \rightarrow 4$, $\theta : 50.0^\circ \rightarrow 22.52^\circ$ (law predicts 22.5°); at $R : 4 \rightarrow 1$ sub-critical, $\theta : 20.0^\circ \rightarrow 43.16^\circ$.

12 Total Spectral Reflection

Theorem 12.1. *Let Σ separate a high-rigidity domain (R_1) from a relaxed domain (R_2), $\kappa = R_1/R_2 > 1$. If the incident angle exceeds $\theta_{\text{crit}} = \arcsin \sqrt{1/\kappa}$, then $\sin^2 \theta_2 = \kappa \sin^2 \theta_1 > 1$ forces $\theta_2 = \frac{\pi}{2} + i \cosh^{-1}(\sqrt{\kappa} \sin \theta_1)$, the normal momentum becomes purely imaginary, the transmitted amplitude decays as $e^{-|p_\perp| w_\perp}$, and the wavefront is totally reflected into the interior of the loaded domain.*

The algebra is elementary from (17) and $\sin(\frac{\pi}{2} + iy) = \cosh y$. *Dynamical verification* [V]: the

super-critical case ($R : 4 \rightarrow 1$, $\theta_1 = 45^\circ > \theta_{\text{crit}} = 30^\circ$) physically reversed in the layer and exited at exactly 45.00° , with all invariants conserved to 10^{-15} (REFRACT-EL-01, Case C).

Remark 12.2 (The Smith-chart dictionary). Define the interface reflection coefficient $r = (p_{\perp,1} - p_{\perp,2}) / (p_{\perp,1} + p_{\perp,2})$ with $p_{\perp} = \sqrt{2R(E - V_0)} \cos \theta$. This is the Cayley–Möbius map of the normal-impedance half-plane onto the unit disk: the engineer’s Smith chart is a coordinate chart on $\text{Gr}(1,2) = \mathbb{CP}^1$, cascaded interfaces act by Möbius (transfer-matrix) maps, the rim $|r| = 1$ is exactly the total reflection regime of Theorem 12.1 (reactive load = evanescent channel), and tapered impedance matching [C] [29] is the mollified layer of §11: an abrupt step reflects, a taper transmits. RF impedance matching is projector-bundle geometry on the smallest Grassmannian.

While expansion is the primary stress-relief mode at cosmic scales, the same architecture admits contraction regimes in which increasing local load drives clustering until trackability is lost (§5); a detailed treatment of structure formation is deferred.

13 Physical Applications

13.1 Cosmological relaxation

A heavy local load raises the non-normal commutator and with it the spectral stress Φ . By Theorem 8.2 the relaxation flow sheds exactly the Henrici budget: increasing relational distance between active nodes drops the off-diagonal couplings and drives $\lim_{|x_i - x_j| \rightarrow \infty} \|[K, K^\dagger]\| = 0$. Cosmic expansion is the macroscopic signature of an overloaded operator network executing structural stress relief, with a finite, exact budget $\Delta_F^2(K_0)$ fixed at the outset. Supervoids are fully relaxed operator sinks ($\Delta K \rightarrow 0$) whose rigidity contrast with filament walls suppresses persistent transport into the interior by §12.

13.2 Quantum topological insulators, upgraded

A two-dimensional topological insulator’s gapped bulk is a loaded operator domain; its physical edge is the interface with a trivial substrate. By §10.2 both sides are gapped ground states, so the interface carries a boundary-trackable channel census obeying the area law, and the entanglement-spectrum/edge-state correspondence [C] [21] makes the protected edge channels precisely the census channels $\nu \in (\epsilon, 1 - \epsilon)$. Backscattering into the bulk is forbidden by the high rigidity contrast through Theorem 12.1. A 330-million-light-year supervoid and a sub-nanometer edge track are protected by the same operator architecture; the container-as-ground-state theorem (Theorem 10.3) is what makes the analogy exact rather than suggestive. The shared engine across the adjacent laboratory literatures— flat-band condensates, charge-neutrality-point transport, nonlinear Hall responses—is the same quantum geometric tensor Q_Π [C] [23, 24, 25].

13.3 Ledger perception geometry: diminishing returns at interfaces

An observer that compares two interface states through the trace-log ledger measures the log-det (Jensen–Bregman) divergence $S(A, B) = \log \det \frac{A+B}{2} - \frac{1}{2} \log \det A - \frac{1}{2} \log \det B$ [C] [28]. On a geodesic family of SPD rigidity operators crossing a wall of contrast κ (COLOR-LEDGER-01, Fig. 4): the Riemannian transport metric is a length metric (accumulated/direct = 1.0000 at every contrast—the control), while the ledger metric $\sqrt{S}$ is locally proportional to it with the exact constant $dS^{1/2}/d\delta_R \rightarrow 1/(2\sqrt{2}) = 0.353553$ (six digits, uniform along the path) yet globally subadditive: accumulated/direct grows $1.005 \rightarrow 1.192$ as $\log \kappa : 1 \rightarrow 8$. Locally Riemannian just-noticeable differences, globally compressed large jumps: precisely the non-Riemannian “diminishing

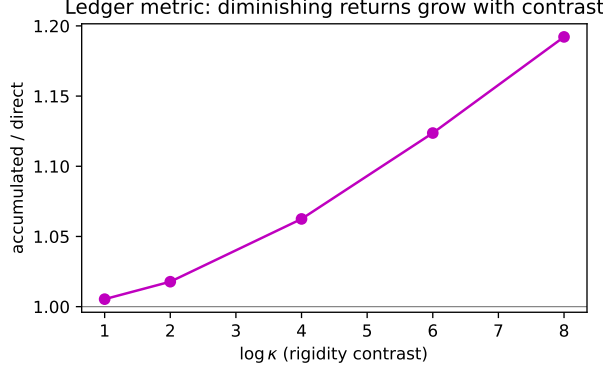


Figure 4: Diminishing returns of the ledger comparison metric (COLOR-LEDGER-01 via FIG-COSMO-01): summed small-step ledger distances exceed the direct ledger distance by a factor growing with the rigidity contrast, while the Riemannian control stays exactly at 1.

returns” geometry measured in human color perception [C] [26, 27]. The corollary for interfaces: a single violent wall crossing is log-compressed in the ledger relative to a graded crossing through the mollified layer—the ledger under-remembers violence relative to adiabatic change, with the layer thickness playing the just-noticeable-step role. We claim the mechanism, not the psychophysics.

14 Open Problems

1. Commit to a horizon cutoff and derive $w(z)$; confront the Hsu obstruction [12] and the DESI DR2 evolving-dark-energy data [13, 14].
2. Derive the $O(1)$ constant α_s from the orientation average of the census density $n_{\text{ch}}(\hat{n})$ (Widom-type integral [17]).
3. Behavior of the relaxation flow (13) at and near defective points (the equilibrium manifold meets the exceptional strata).
4. Lorentzian signature and causal structure from the non-normal sector.
5. Uniqueness structure of the equilibrium selected by the global projector-bundle constraints (when does the manifold of Corollary 8.3 reduce to $-\Lambda I$?).
6. Physical consequences of the corrected color dynamics (7).
7. Normalizable spectral densities realizing the chiral continuum limit of Theorem 6.2.

A Off-Diagonal Projector Geometry and the Intrinsic QGT

Let $\text{Gr}(k, \mathcal{H})$ be the Grassmannian of rank- k orthogonal projectors $\Pi^\dagger = \Pi = \Pi^2$, $\text{Tr} \Pi = k$. Differentiating $\Pi^2 = \Pi$ gives the tangent constraint $V = \Pi V(I - \Pi) + (I - \Pi)V\Pi$: tangent vectors are strictly off-diagonal. On this tangent space define $Q_\Pi(V, W) = \text{Tr}(\Pi V W)$, which decomposes as $Q_\Pi = g_\Pi + \frac{i}{2}\Omega_\Pi$ with intrinsic metric $g_\Pi(V, W) = \frac{1}{2}\text{Tr}(V W)$ and Berry curvature $\Omega_\Pi(V, W) = -i\text{Tr}(\Pi[V, W])$. *Verification* [V]: the decomposition holds to 2×10^{-16} on random off-diagonal Hermitian tangent vectors (gate G4).

Spectral realization (cross-gap transitions only). For $V = \delta\Pi_k$ generated by a self-adjoint $\delta\Sigma$, $\langle m|V|n\rangle = \langle m|\delta\Sigma|n\rangle/(\lambda_m - \lambda_n)$ across the gap and zero otherwise: metric and curvature are cross-gap objects weighted by inverse gaps.

Sharp metric–curvature inequality. $|\Omega_\Pi(V, W)| \leq 2\sqrt{g_\Pi(V, V)g_\Pi(W, W) - g_\Pi(V, W)^2} \leq 2\sqrt{g_\Pi(V, V)g_\Pi(W, W)}$ curvature is suppressed by alignment of admissible channels. Verified numerically in G4.

Maximally isotropic baseline. An operator commuting with every rank- k projector commutes with every orthogonal projector onto every k -dimensional subspace, hence with every rank-one projector obtained from their differences and intersections, hence is a scalar: $K_0 = -\Lambda I$ is the unique zero-commutator baseline compatible with the whole bundle. Any off-diagonal perturbation generates both a commutator and a nonzero metric.

B Structural Decomposition and Relaxation

Tripartite decomposition. $K = A + D_0 + \varepsilon E$: transport (Hermitian, real spectrum), obstruction (symmetric Hessians), non-normal generator (εE , the sole source of non-normality, skew to leading order in the obstruction gradients that generate it). When $\varepsilon E = 0$ the operator is normal.

Spectral stress and flow. $\Phi(K) = \frac{1}{2}\|[K, K^\dagger]\|_F^2$ vanishes iff K is normal; its steepest descent is the flow (13) of Theorem 8.1, with descent rate $-4\|[C, K]\|_F^2$ and the exact dissipation identity of Theorem 8.2. The flow is isospectral, so the endpoint is a normal operator carrying the initial spectrum; the equilibrium set is the unitary orbit of Corollary 8.3. The isotropic tare state $-\Lambda I$ is the distinguished equilibrium of the degenerate-spectrum sector and, by the commutant argument of Appendix A, the unique baseline compatible with the entire persistent projector bundle. In the tripartite language, relaxation is the dissipation of the non-normal generator, $\varepsilon E \rightarrow$ alignment, at frozen spectral content.

C Explicit Transport Prototypes

Three-body prototype. On the Kendall shape sphere the intrinsic transport operator is $A_{\text{intr}} = J^\top P_H T(w) J$ with $T(w) = \text{diag}(w_1^2, w_2^2, w_3^2)$. Near the Euler equator the smallest singular value obeys $\sigma_{\min}(A_{\text{intr}}) \sim \cos^2 \theta$ and the relational rigidity $R(w) \sim 1/\cos^4 \theta$; the mechanism is the 2×2 symmetric-PSD eigenvalue algebra ($\det \sim \epsilon^2$, $\text{trace} = O(1)$, so $\lambda_{\min} \sim \det/\text{trace}$), established in the companion three-body audit.

Defective interface subsystem. At interface coalescence the reduced block $K_{\text{red}} = bI + N$, $N^2 = 0$, is a size-2 Jordan block; the resolvent obeys $\|R(\lambda)\| \approx |\mu|/|\lambda - b|^2$, so the ε -pseudospectrum boundary expands as the square-root deformation $|\lambda - b| = O(\varepsilon^{1/2})$, and the invariant-subspace lift transfers the polynomial transient growth $\|e^{-Jt}\| \leq (1 + |\mu|t)e^{-bt}$ to the global operator. Conical rank collapse at $\mu \rightarrow \mu_c$ suppresses transverse velocity as $v_\perp \propto 1/\sqrt{2\lambda}$, flattening transport cones parallel to the interface—the microscopic realization of the directional filtering of §11–12.

Grassmannian lift. The Riesz projector $\Pi_k = \frac{1}{2\pi i} \oint_\Gamma (zI - K)^{-1} dz$ lifts the trajectory to $\text{Gr}(k, \mathcal{H})$; the persistence overlap $C(\Delta) = \text{Tr}(\Pi(t)\Pi(t + \Delta))$ quantifies how admissible channels outlive individual eigenvector reorientation.

D Chiral Collapse, Holonomy Phase, and the Two Constants

Holonomy phase and Catalan’s constant $[T] + [V]$. The flat-torus phase form $d\Phi = d\arctan(Y/X)$ along the sweep $X = 1$, $Y = xt$ accumulates $\Phi(x) = \arctan x$, and integration against the scale-

invariant measure dx/x gives

$$G = \int_0^1 \frac{\arctan x}{x} dx = 0.9159655941 \dots \quad (18)$$

verified to 30 digits (COSMO-CATALAN-01).

Continuum ledger [T]+[V]. With the chiral skew normalization of Theorem 6.2, the normalized ledger converges to $\frac{1}{2} \int \log(1 + \lambda^2) \rho(\lambda) d\lambda = \pi^2/12$ under the density $\rho(\lambda) = \arctan(1/\lambda)/(\pi\lambda)$; the underlying closed form is $\frac{2}{\pi} \int_0^\infty \lambda^{-1} \log(1 + \lambda^2) \arctan(1/\lambda) d\lambda = \pi^2/6$. The holonomy constant and the ledger constant are distinct invariants of distinct objects.

E Scripts

Every computational gate cited in this paper is generated by the scripts below, listed in full and in ASCII. Each script carries its name in a header comment. They require only `numpy`, `scipy`, `mpmath`, and `matplotlib`.

E.1 COSMO-GRAD-01 (gradient identification; descent test; isospectrality)

```
# SCRIPT: COSMO-GRAD-01 (gradient of Phi(K)=0.5*||[K,K+]||_F^2 : paper's [[C],K+] vs [[C],K])
import numpy as np
rng = np.random.default_rng(7)
def comm(A,B): return A@B - B@A
def C_of(K): return comm(K, K.conj().T)
def Phi(K):
    C = C_of(K); return 0.5*np.real(np.trace(C @ C.conj().T))

print("=== TEST A: which candidate matches the finite-difference directional derivative ===")
for seed in range(4):
    r = np.random.default_rng(seed)
    n = 6
    K = r.standard_normal((n,n)) + 1j*r.standard_normal((n,n))
    D = r.standard_normal((n,n)) + 1j*r.standard_normal((n,n))
    eps = 1e-6
    fd = (Phi(K+eps*D) - Phi(K-eps*D))/(2*eps)
    C = C_of(K)
    G_paper = comm(C, K.conj().T) # [[K,K+],K+]
    G_mine = comm(C, K) # [[K,K+],K]
    pred_paper = 2*np.real(np.trace(G_paper.conj().T @ D))
    pred_mine = 2*np.real(np.trace(G_mine.conj().T @ D))
    print(f"seed {seed}: FD={fd:+.6f} pred[[C],K+]={pred_mine:+.6f} pred[[C],K+]={pred_paper:+.6f}")

print()
print("=== TEST B: paper's descent identity dPhi/dt = -4*||[[C],K+]||^2 along the paper's own flow ===")
for seed in range(4):
    r = np.random.default_rng(100+seed)
    n = 6
    K = r.standard_normal((n,n)) + 1j*r.standard_normal((n,n))
    C = C_of(K)
    F = -2*comm(C, K.conj().T) # paper's flow direction
    G_true = comm(C, K) # true gradient (from Test A)
    actual = 2*np.real(np.trace(G_true.conj().T @ F)) # actual dPhi/dt along F
    eps = 1e-6
    fd_actual = (Phi(K+eps*F) - Phi(K-eps*F))/(2*eps) # same thing, pure finite difference
    claimed = -4*np.real(np.trace(comm(C,K.conj().T).conj().T @ comm(C,K.conj().T)))
    print(f"seed {100+seed}: actual dPhi/dt = {fd_actual:+.4f} paper's claimed value = {claimed:+.4f}")

print()
print("=== TEST C: 2x2 nilpotent K=[[0,1],[0,0]] -- clean counterexample ===")
K = np.array([[0,1],[0,0]], dtype=complex)
C = C_of(K)
```

```

F = -2*comm(C, K.conj().T)
eps=1e-6
fd = (Phi(K+eps*F)-Phi(K-eps*F))/(2*eps)
claimed = -4*np.real(np.trace(comm(C,K.conj().T).conj().T @ comm(C,K.conj().T)))
print(f"K nilpotent: actual dPhi/dt along paper flow = {fd:+.6f}; paper's formula predicts {claimed:+.6f}")

print()
print("=== TEST D: the TRUE gradient flow dK/dt=-2[[C],K] is isospectral (eigenvalues frozen) ===")
r = np.random.default_rng(42); n=5
K = r.standard_normal((n,n)) + 1j*r.standard_normal((n,n))
ev0 = np.sort_complex(np.linalg.eigvals(K))
h = 2e-4
Kt = K.copy()
for step in range(40000):
    # RK4 on dK/dt = -2[C(K),K]
    def f(M): return -2*comm(C_of(M), M)
    k1=f(Kt); k2=f(Kt+0.5*h*k1); k3=f(Kt+0.5*h*k2); k4=f(Kt+h*k3)
    Kt = Kt + (h/6)*(k1+2*k2+2*k3+k4)
ev1 = np.sort_complex(np.linalg.eigvals(Kt))
print("eigenvalue drift after long integration:", np.max(np.abs(ev1-ev0)))
print("residual non-normality ||[K,K+]|| :", np.linalg.norm(C_of(Kt)))
print("initial non-normality :", np.linalg.norm(C_of(K)))
print("distance of final K from -Lam*I? final eigenvalues:", np.round(ev1,3))

```

E.2 COSMO-BIORTHO-02 (dissipation identity; alignment; dual flow)

```

# SCRIPT: COSMO-BIORTHO-02 (co-vector/vector alignment along the corrected flow; Henrici dissipation identity; paper-flow control)
import numpy as np
from scipy.linalg import eig
np.set_printoptions(precision=4, suppress=False)

def comm(A,B): return A@B - B@A
def C_of(K): return comm(K, K.conj().T)
def Phi(K):
    C = C_of(K); return 0.5*np.real(np.trace(C @ C.conj().T))
def kappas(K):
    w, vl, vr = eig(K, left=True, right=True) # columns unit-norm
    idx = np.lexsort((w.imag, w.real))
    w, vl, vr = w[idx], vl[:,idx], vr[:,idx]
    kap = np.array([1.0/abs(vl[:,i].conj() @ vr[:,i]) for i in range(len(w))])
    return w, kap, vr

print("=== TEST 1: exact identity d/dt ||K||_F^2 = -8*Phi(K) along corrected flow (finite difference) ===")
for seed in range(5):
    r = np.random.default_rng(seed); n=6
    K = r.standard_normal((n,n)) + 1j*r.standard_normal((n,n))
    F = -2*comm(C_of(K), K)
    eps = 1e-7
    fd = (np.linalg.norm(K+eps*F)**2 - np.linalg.norm(K-eps*F)**2)/(2*eps)
    print(f"seed {seed}: FD d||K||^2/dt = {fd:+.6f} -8*Phi = {-8*Phi(K):+.6f} ratio = {fd/(-8*Phi(K)):.9f}")

print()
print("=== TEST 2: trajectory -- eigenvalues frozen, kappas -> 1, Henrici budget closes ===")
r = np.random.default_rng(11); n=6
K0 = r.standard_normal((n,n)) + 1j*r.standard_normal((n,n))
w0, kap0, vr0 = kappas(K0)
S2 = np.sum(np.abs(w0)**2) # conserved spectral energy
henrici0 = np.linalg.norm(K0)**2 - S2
print(f"t=0: Henrici^2 = {henrici0:.6f} Phi = {Phi(K0):.4f}")
print(f"t=0 kappas: ", np.round(kap0,4))
lead0 = np.argmax(np.abs(w0)) # leading = largest |lambda|
Kt = K0.copy(); h = 5e-5; t = 0.0
checkpoints = [0.001,0.005,0.02,0.1,0.5,2.0]
ci = 0; log = []
def f(M): return -2*comm(C_of(M), M)
while ci < len(checkpoints):
    k1=f(Kt); k2=f(Kt+0.5*h*k1); k3=f(Kt+0.5*h*k2); k4=f(Kt+h*k3)

```



```

Kt = Kt + (h/6)*(k1+2*k2+2*k3+k4); t += h
if t >= checkpoints[ci]:
    w, kap, vr = kappas(Kt)
    hen = np.linalg.norm(Kt)**2 - S2
    lead = np.argmax(np.abs(w))
    ang = np.degrees(np.arccos(min(1.0, abs(vr0[:,lead0].conj() @ vr[:,lead])))
    log.append((t, np.max(np.abs(np.sort_complex(w)-np.sort_complex(w0))), hen, Phi(Kt), np.max(kap), kap[lead], ang))
    ci += 1
print(f"{'t':>7} {'eig drift':>10} {'Henrici^2':>10} {'Phi':>10} {'max kappa':>10} {'kappa_lead':>10} {'R1 rotation(deg)':>16}")
for row in log:
    print(f"{row[0]:7.3f} {row[1]:10.2e} {row[2]:10.4f} {row[3]:10.4f} {row[4]:10.5f} {row[5]:10.5f} {row[6]:16.2f}")
print("limit spectrum (should equal initial):", np.round(np.sort_complex(np.linalg.eigvals(Kt)),3))
print("initial spectrum", np.round(np.sort_complex(w0),3))

print()
print("=== TEST 3: budget check -- integral of 8*Phi dt vs Henrici^2 dissipated ===")
Kt = K0.copy(); h=5e-5; acc=0.0
for step in range(int(2.0/h)):
    acc += 8*Phi(Kt)*h
    k1=f(Kt); k2=f(Kt+0.5*h*k1); k3=f(Kt+0.5*h*k2); k4=f(Kt+h*k3)
    Kt = Kt + (h/6)*(k1+2*k2+2*k3+k4)
print(f"integral 8*Phi dt over [0,2] = {acc:.6f}")
print(f"Henrici^2 drop ||K0||^2-||K(2)||^2 = {np.linalg.norm(K0)**2 - np.linalg.norm(Kt)**2:.6f}")

print()
print("=== TEST 4: the paper's flow dK/dt=-2[[C],[K]] -- spectrum NOT preserved; where does it go? ===")
def fp(M): return -2*comm(C_of(M), M.conj().T)
Kt = K0.copy(); h=2e-5; t=0
for tstop in [0.01, 0.05, 0.2, 1.0]:
    while t < tstop:
        k1=fp(Kt); k2=fp(Kt+0.5*h*k1); k3=fp(Kt+0.5*h*k2); k4=fp(Kt+h*k3)
        Kt = Kt + (h/6)*(k1+2*k2+2*k3+k4); t += h
        if not np.isfinite(np.linalg.norm(Kt)) or np.linalg.norm(Kt) > 1e8:
            print(f"t={t:.4f}: BLOW-UP, ||K|| = {np.linalg.norm(Kt):.2e}"); break
    else:
        w = np.sort_complex(np.linalg.eigvals(Kt))
        print(f"t={t:.3f}: ||K||={np.linalg.norm(Kt):9.3f} Phi={Phi(Kt):10.3f} eig drift={np.max(np.abs(w-np.sort_complex(w0)))}
        continue
    break

```

E.3 COSMO-CATALAN-01 (chiral identity; the two constants; density mass)

```

# SCRIPT: COSMO-CATALAN-01 (chiral ledger identity 6.12 + Catalan integral 13.13 + density normalization)
import numpy as np
from mpmath import mp, quad, log, atan, pi, catalan, inf, mpf
mp.dps = 30

print("=== TEST A: algebraic identity (6.12) on an explicitly chiral matrix ===")
r = np.random.default_rng(3)
m = 5
B = 0.15*(r.standard_normal((m,m)) + 1j*r.standard_normal((m,m)))
X = np.block([[np.zeros((m,m)), B],[B.conj().T, np.zeros((m,m))]]) # anticommutes with diag(I,-I)
ev = np.linalg.eigvalsh(X)
print("spectrum symmetric? max|ev + reversed(ev)| =", np.max(np.abs(ev + ev[::-1])))
Z_true = np.sum(np.log(1+ev)) # Tr ln(I+X) = the actual ledger from Thm 6.1's series
Z_paper = np.sum(np.log(1+ev**2)) # paper's (6.12): Tr ln(I+X^2)
Z_half = 0.5*np.sum(np.log(1-ev**2)) # candidate correction: (1/2) Tr ln(I-X^2)
print(f"Tr ln(I+X) = {Z_true:+.12f} (ground truth)")
print(f"paper Tr ln(I+X^2) = {Z_paper:+.12f}")
print(f"1/2 Tr ln(I-X^2) = {Z_half:+.12f}")

print()
print("=== TEST B: integral identity (13.13): (2/pi) * int_0^inf ln(1+l^2) arctan(1/l)/l dl vs G ===")
I = (2/pi)*quad(lambda l: log(1+l**2)*atan(1/l)/l, [0, 1, inf])
print("integral value =", I)
print("Catalan G =", +catalan)
print("ratio I/G =", I/catalan)

```

```

print("I/(pi) ? ", I/pi, " I - pi*ln2 ? ", I - pi*log(2))
# sanity: the clean identity the paper DOES have right:
print("int_0^1 arctan(x)/x dx =", quad(lambda x: atan(x)/x, [0,1]), " (should be G)")

print()
print("=== TEST C: is rho(l)=arctan(1/l)/(pi*l) a probability density? ===")
for L in [10, 100, 1000, 10000]:
    tot = (2/pi)*quad(lambda l: atan(1/l)/l, [mpf('1e-8'), 1, L])
    print(f"int_{{-{{L}}}}^{{{{L}}}} rho = {float(tot):.4f}")

```

E.4 RANK-COUNT-01 (area law of the census, with gapless control)

```

# SCRIPT: RANK-COUNT-01 (boundary-trackable projector census: area law from the gap/trackability hypothesis)
import numpy as np

def band_projector(H, tol=0.0):
    ev, V = np.linalg.eigh(H)
    occ = ev < tol
    return V[:, occ] @ V[:, occ].conj().T, ev

def census(P, idx, eps=1e-3):
    C = P[np.ix_(idx, idx)]
    nu = np.linalg.eigvalsh(C)
    return int(np.sum((nu > eps) & (nu < 1-eps)))

print("=== 1D chain, N_track(eps=1e-3) vs region length L ===")
Ntot = 1200; th = 0.4
def H1d(m):
    H = np.zeros((Ntot, Ntot), complex)
    ph = np.exp(1j*th/Ntot)
    for j in range(Ntot):
        H[j, (j+1)%Ntot] = -ph; H[(j+1)%Ntot, j] = -np.conj(ph)
        H[j, j] = m*(-1)**j
    return H
for m, tag in [(0.5, "GAPPED m=0.5 (P1: expect flat)"), (0.0, "GAPLESS m=0 (P2: expect ~log L growth)"]]:
    P, _ = band_projector(H1d(m))
    counts = [(L, census(P, np.arange(L))) for L in [50, 100, 200, 400]]
    print(f" {tag}: " + " ".join(f"L={L}: N={N}" for L, N in counts))

print()
print("=== 1D gap dial at fixed L=200 (P5: N grows as gap shrinks) ===")
for m in [1.0, 0.5, 0.2, 0.1, 0.05]:
    P, _ = band_projector(H1d(m))
    print(f" m={m:5.2f} N={census(P, np.arange(200))}")

print()
print("=== 2D square lattice 44x44, region LxL (P3/P4) ===")
Lx = Ly = 44; Nsites = Lx*Ly
def idx2(x, y): return (x%Lx)*Ly + (y%Ly)
def H2d(m):
    H = np.zeros((Nsites, Nsites), complex)
    phx, phy = np.exp(1j*0.37/Lx), np.exp(1j*0.83/Ly)
    for x in range(Lx):
        for y in range(Ly):
            i = idx2(x, y)
            H[i, idx2(x+1, y)] += -phx; H[idx2(x+1, y), i] += -np.conj(phx)
            H[i, idx2(x, y+1)] += -phy; H[idx2(x, y+1), i] += -np.conj(phy)
            H[i, i] = m*(-1)**(x+y)
    return H
for m, tag in [(1.0, "GAPPED m=1.0"), (0.0, "GAPLESS m=0")]:
    P, _ = band_projector(H2d(m))
    Ls = [6, 10, 14, 18, 22]; Ns = []
    for L in Ls:
        region = [idx2(x, y) for x in range(L) for y in range(L)]
        Ns.append(census(P, np.array(region)))
    slope = np.polyfit(np.log(Ls), np.log(Ns), 1)[0]
    perim = " ".join(f"L={L}: N={N} (N/4L={N/(4*L):.3f})" for L, N in zip(Ls, Ns))

```

```

print(f" {tag}: {perim}")
print(f"    log-log slope N vs L = {slope:.3f}    ({'area-law' if slope < 1.15 else 'violation'} territory)")

```

E.5 CONTAINER-GROUND-01 (exchange symmetry; shape independence)

```

# SCRIPT: CONTAINER-GROUND-01 (his hypothesis: the container subspace is a ground state on equal footing; shape-independence)
import numpy as np
Lx = Ly = 44; Nsites = Lx*Ly
def idx2(x,y): return (x%Lx)*Ly + (y%Ly)
H = np.zeros((Nsites, Nsites), complex)
phx, phy = np.exp(1j*0.37/Lx), np.exp(1j*0.83/Ly)
m = 1.0
for x in range(Lx):
    for y in range(Ly):
        i = idx2(x,y)
        H[i, idx2(x+1,y)] += -phx; H[idx2(x+1,y), i] += -np.conj(phx)
        H[i, idx2(x,y+1)] += -phy; H[idx2(x,y+1), i] += -np.conj(phy)
        H[i,i] = m*(-1)**(x+y)
ev, V = np.linalg.eigh(H)
Vocc = V[:, ev < 0]
P = Vocc @ Vocc.conj().T
eps = 1e-3

def region_square(L, x0=4, y0=4):
    return [idx2(x0+x, y0+y) for x in range(L) for y in range(L)]
def region_diamond(r, cx=22, cy=22):
    return [idx2(x,y) for x in range(Lx) for y in range(Ly) if abs(x-cx)+abs(y-cy) <= r]
def boundary_sites(A):
    S = set(A); cnt = 0
    for i in A:
        x, yy = divmod(i, Ly)
        nbrs = [idx2(x+1,yy), idx2(x-1,yy), idx2(x,yy+1), idx2(x,yy-1)]
        if any(nb not in S for nb in nbrs): cnt += 1
    return cnt
def mid_spectrum(Mat):
    nu = np.linalg.eigvalsh(Mat)
    return np.sort(nu[nu > eps] & (nu < 1-eps)])

print("=== CLAIM 1: state/container swap symmetry (square L=14) ===")
A = np.array(region_square(14))
nu_state_in_container = mid_spectrum(P[np.ix_(A,A)])
D = np.zeros(Nsites); D[A] = 1.0
M2 = Vocc.conj().T @ (D[:,None]*Vocc)
nu_container_in_state = mid_spectrum(M2)
print(f"channels either way: {len(nu_state_in_container)} vs {len(nu_container_in_state)}")
print(f"max |difference| of matched spectra: {np.max(np.abs(nu_state_in_container - nu_container_in_state)):.2e}")
print(f"(H_A = I - 2*1_A: flat bands at +-1, gap 2 -- the container IS a gapped ground state.)")

print()
print("=== CLAIM 2: container-shape independence (N per boundary site) ===")
for name, A in [("square L=14", region_square(14)), ("square L=18", region_square(18)),
                ("diamond r=10", region_diamond(10)), ("diamond r=13", region_diamond(13))]:
    A = np.array(A)
    N = len(mid_spectrum(P[np.ix_(A,A)]))
    b = boundary_sites(A)
    print(f" {name:14s}: |A|={len(A):4d}  boundary sites={b:3d}  N={N:3d}  N/boundary = {N/b:.3f}")

```

E.6 COLOR-LEDGER-01 (ledger diminishing returns vs Riemannian control)

```

# SCRIPT: COLOR-LEDGER-01 (diminishing returns of the log-det ledger metric vs the Riemannian transport metric on SPD rigidity)
import numpy as np
from scipy.linalg import fractional_matrix_power as fmp, sqrtm
rng = np.random.default_rng(2)
d = 5

def delta_R(A,B):
    # affine-invariant Riemannian distance (length metric -- the control)

```

```

    Ai = fmp(A,-0.5)
    ev = np.linalg.eigvalsh(Ai@B@Ai)
    return np.sqrt(np.sum(np.log(ev)**2))
def dS(A,B):
    # ledger metric: sqrt of Jensen-Bregman log-det divergence
    s = np.linalg.slogdet((A+B)/2)[1] - 0.5*np.linalg.slogdet(A)[1] - 0.5*np.linalg.slogdet(B)[1]
    return np.sqrt(max(s,0.0))

# relaxed side of the wall: near-identity SPD
Q = rng.standard_normal((d,d)); A0 = np.eye(d) + 0.05*(Q+Q.T) @ (Q+Q.T).T/ d
# loaded side: rigidity contrast kappa along a random anisotropy frame
U,_ = np.linalg.qr(rng.standard_normal((d,d)))

print(f"{'log kappa':>9} {'direct dR':>10} {'accum dR':>10} {'ratio dR':>9} | {'direct dS':>10} {'accum dS':>10} {'ratio dS':>9}")
for lk in [1,2,4,6,8]:
    spread = np.linspace(-lk/2, lk/2, d)
    A1 = U @ np.diag(np.exp(spread)) @ U.T          # SPD, condition number e^lk
    # geodesic path A(t) = A0^-1/2 (A0^-1/2 A1 A0^-1/2)^t A0^-1/2, fine discretization
    A0h = sqrtm(A0).real; A0hi = np.linalg.inv(A0h)
    Mid = A0hi @ A1 @ A0hi
    n = 400
    path = [A0h @ fmp(Mid, t).real @ A0h for t in np.linspace(0,1,n+1)]
    accR = sum(delta_R(path[k], path[k+1]) for k in range(n))
    accS = sum(dS(path[k], path[k+1]) for k in range(n))
    dirR = delta_R(A0, A1); dirS = dS(A0, A1)
    print(f"{lk:9d} {dirR:10.4f} {accR:10.4f} {accR/dirR:9.4f} | {dirS:10.4f} {accS:10.4f} {accS/dirS:9.4f}")

print()
print("local calibration (step size -> 0): dS/dR on one small step at three points along the path:")
for t in [0.1, 0.5, 0.9]:
    Aa = path[int(t*n)]; Ab = path[int(t*n)+1]
    print(f"    t={t}: dS/dR = {dS(Aa,Ab)/delta_R(Aa,Ab):.6f}    (1/(2*sqrt2) = {1/(2*np.sqrt(2)):.6f})")

```

E.7 REFRACT-EL-01 (dynamical refraction and total reflection)

```

# SCRIPT: REFRACT-EL-01 (integrate the actual EL dynamics through the mollified rigidity layer; test  $R \sin^2$  vs  $R \sin$ )
import numpy as np
eps_layer = 0.05
def make_R(R1, R2):
    Rf = lambda w: (R1+R2)/2 + (R2-R1)/2*np.tanh(w/eps_layer)
    dRf = lambda w: (R2-R1)/2/np.cosh(w/eps_layer)**2/eps_layer
    return Rf, dRf

def run(R1, R2, th1_deg, tmax=6.0, h=2e-5):
    Rf, dRf = make_R(R1, R2)
    th1 = np.radians(th1_deg)
    w, vpar, vper = -1.0, np.sin(th1), np.cos(th1)
    p0 = Rf(w)*vpar; E0 = 0.5*Rf(w)*(vpar**2+vper**2)
    def f(s):
        w, vpar, vper = s
        R, dR = Rf(w), dRf(w)
        return np.array([vper, -dR*vper*vpar/R, dR*(vpar**2 - vper**2)/(2*R)])
    s = np.array([w, vpar, vper]); t = 0.0
    while t < tmax:
        k1=f(s); k2=f(s+0.5*h*k1); k3=f(s+0.5*h*k2); k4=f(s+h*k3)
        s = s + (h/6)*(k1+2*k2+2*k3+k4); t += h
        if s[0] > 1.0 or (s[0] < -1.0 and s[2] < 0): break
    w, vpar, vper = s
    R = Rf(w)
    th_out = np.degrees(np.arcsin(abs(vpar)/np.hypot(vpar,vper)))
    outcome = "TRANSMITTED" if w > 0 else "REFLECTED"
    p_err = abs(R*vpar - p0)/abs(p0)
    E_err = abs(0.5*R*(vpar**2+vper**2) - E0)/E0
    sin2_in, sin2_out = R1*np.sin(th1)**2, R*(abs(vpar)/np.hypot(vpar,vper))**2
    sin_in, sin_out = R1*np.sin(th1), R*(abs(vpar)/np.hypot(vpar,vper))
    print(f"R1={R1} -> R2={R2}, th1={th1_deg:5.1f}deg: {outcome}, th_out={th_out:6.2f}deg | "
          f"p_par drift {p_err:.1e}, E drift {E_err:.1e}")
    print(f"    R sin^2: in {sin2_in:.6f} out {sin2_out:.6f} (rel diff {abs(sin2_in-sin2_out)/sin2_in:.1e})")

```

```

f" | R sin: in {sin_in:.4f} out {sin_out:.4f} (rel diff {abs(sin_in-sin_out)/sin_in:.2f})")

print("Case A: low -> high rigidity, transmit and bend toward normal")
run(1.0, 4.0, 50.0)
print("Case B: high -> low, SUB-critical (theta_crit = arcsin sqrt(1/4) = 30 deg)")
run(4.0, 1.0, 20.0)
print("Case C: high -> low, SUPER-critical -> total reflection expected")
run(4.0, 1.0, 45.0)

```

E.8 READ2-GATES-01 (zeta series; chiral repair; Lipschitz; QGT; back-reaction; SU(3))

```

# SCRIPT: READ2-GATES-01 (remaining read-#2 verification gates for the cosmology paper)
import numpy as np
rng = np.random.default_rng(9)
def comm(A,B): return A@B - B@A

print("G1: Theorem 6.1 zeta-ledger series (traceless dK, eta-Lam=2)")
M = 10
dK = rng.standard_normal((M,M)) + 1j*rng.standard_normal((M,M))
dK -= np.trace(dK)/M*np.eye(M) # traceless
dK *= 0.8/np.linalg.norm(dK,2) # ||X|| < 1
g = 2.0; X = dK/g
Z_true = np.linalg.slogdet(g*np.eye(M) + dK)[1]
Z_series = M*np.log(g) + sum((-1)**(k-1)/k)*np.trace(np.linalg.matrix_power(X,k)).real for k in range(2,60))
print(f" direct log|det| = {Z_true:.12f} formula (6.3) = {Z_series:.12f} diff = {abs(Z_true-Z_series):.1e}")

print("G2: chiral ledger repair -- anti-Hermitian X gives Z = (1/2)Tr ln(I+Y^2)")
m2 = 5
B = 0.3*(rng.standard_normal((m2,m2)) + 1j*rng.standard_normal((m2,m2)))
Y = np.block([[np.zeros((m2,m2)), B], [B.conj().T, np.zeros((m2,m2))]]) # Hermitian, chiral
X = 1j*Y # anti-Hermitian, chiral
Z_true = np.linalg.slogdet(np.eye(2*m2) + X)[1]
lam = np.linalg.eigvalsh(Y)
Z_half = 0.5*np.sum(np.log(1+lam**2))
print(f" log|det(I+X)| = {Z_true:.12f} (1/2)Tr ln(I+Y^2) = {Z_half:.12f} diff = {abs(Z_true-Z_half):.1e}")
print(" -> continuum value under the paper's own density: (1/2)*(pi^2/6) = pi^2/12; Catalan stays with the holonomy integral")

print("G3: Theorem 7.1 Lipschitz bound, numeric (Riesz projectors by contour quadrature)")
n = 6
A = np.diag([1.0,1.2,2.6,3.1,3.7,4.3]) + 0.1*rng.standard_normal((n,n))
c0, r = 1.1+0j, 0.7
ths = np.linspace(0, 2*np.pi, 800, endpoint=False)
zs = c0 + r*np.exp(1j*ths)
sup_res = max(np.linalg.norm(np.linalg.inv(z*np.eye(n)-A), 2) for z in zs)
evs = np.linalg.eigvals(A)
outside = evs[np.abs(evs-c0) > r]
dgap = min(abs(ev-c0) for ev in outside) - r
C_G = sup_res*dgap
eps = 0.3*dgap/C_G
E = rng.standard_normal((n,n)); E *= eps/np.linalg.norm(E,2)
def riesz(Mt):
    S = np.zeros((n,n), complex)
    for z in zs: S += np.linalg.inv(z*np.eye(n)-Mt)*(1j*r*np.exp(1j*np.angle(z-c0)))
    return S*(2*np.pi/len(zs))/(2j*np.pi)
Pi, Pip = riesz(A), riesz(A+E)
lhs = np.linalg.norm(Pip-Pi, 2)
ell = 2*np.pi*r
K_G = ell*C_G**2/(2*np.pi*dgap*(dgap-eps*C_G))
print(f" ||Pi'-Pi|| = {lhs:.4f} <= K_Gamma*eps = {K_G*eps:.4f} -> {'BOUND HOLDS' if lhs <= K_G*eps else 'VIOLATED'} (Tr P

print("G4: Appendix A conventions -- Q = g + (i/2)Omega with g=(1/2)Tr(VW), Omega=-i Tr(Pi[V,W]); and the metric-curvature inequ
k, nn = 2, 6
Q0,_ = np.linalg.qr(rng.standard_normal((nn,nn)) + 1j*rng.standard_normal((nn,nn)))
Pi = Q0[:, :k] @ Q0[:, :k].conj().T
def tangent():
    Vr = rng.standard_normal((nn,nn)) + 1j*rng.standard_normal((nn,nn)); Vr = (Vr+Vr.conj().T)/2

```

```

    Qp = np.eye(nn)-Pi
    return Pi@Vr@Qp + Qp@Vr@Pi
V, W = tangent(), tangent()
Q_VW = np.trace(Pi@V@W)
gVW = 0.5*np.trace(V@W).real
Om = (-1j*np.trace(Pi@comm(V,W))).real
print(f"   Tr(PiVW) = {Q_VW:.6f}   g + (i/2)Om = {gVW + 0.5j*Om:.6f}   diff = {abs(Q_VW-(gVW+0.5j*Om)):.1e}")
gVV, gWW = 0.5*np.trace(V@V).real, 0.5*np.trace(W@W).real
rhs = 2*np.sqrt(max(gVV*gWW - gVW**2, 0))
print(f"   |Omega| = {abs(Om):.6f}   <=   2*sqrt(gg-g^2) = {rhs:.6f}   -> {'HOLDS' if abs(Om)<=rhs+1e-12 else 'VIOLATED'}")

print("G5: corrected back-reaction linearization (Sec 3 repair): dF[V] = -2([V,K+]+[K,V+],K) + [[K,K+],V])")
K = rng.standard_normal((5,5)) + 1j*rng.standard_normal((5,5))
V = rng.standard_normal((5,5)) + 1j*rng.standard_normal((5,5))
def F(Kk): return -2*comm(comm(Kk,Kk.conj().T), Kk)
h = 1e-6
fd = (F(K+h*V) - F(K-h*V))/(2*h)
dC = comm(V,K.conj().T) + comm(K,V.conj().T)
formula = -2*(comm(dC,K) + comm(comm(K,K.conj().T), V))
print(f"   ||FD - formula|| / ||FD|| = {np.linalg.norm(fd-formula)/np.linalg.norm(fd):.1e}")

print("G6: Eq (4.9) SU(3) commutator, numeric")
# Gell-Mann matrices
l = np.zeros((8,3,3), complex)
l[0][0,1]=l[0][1,0]=1; l[1][0,1]=-1j; l[1][1,0]=1j; l[2][0,0]=1; l[2][1,1]=-1
l[3][0,2]=l[3][2,0]=1; l[4][0,2]=-1j; l[4][2,0]=1j; l[5][1,2]=l[5][2,1]=1
l[6][1,2]=-1j; l[6][2,1]=1j; l[7]=np.diag([1,1,-2])/np.sqrt(3)
f = np.zeros((8,8,8))
for a in range(8):
    for b in range(8):
        cab = comm(l[a],l[b])
        for c in range(8): f[a,b,c] = (np.trace(cab@l[c])/(4j)).real
psi = rng.standard_normal(8) + 1j*rng.standard_normal(8)
A49 = sum(psi[a]*l[a] for a in range(8))
lhs = comm(A49, A49.conj().T)
rhs = -2*sum(f[a,b,c]*np.imag(psi[a]*np.conj(psi[b]))*l[c] for a in range(8) for b in range(8) for c in range(8))
print(f"   ||[A,A+] - RHS(4.9)|| = {np.linalg.norm(lhs-rhs):.1e}   (traceless: {abs(np.trace(lhs)):.1e})")

```

E.9 FIG-COSMO-01 (all figures; corrected color-flow gate)

```

# SCRIPT: FIG-COSMO-01 (all v2 figures regenerated from the gated computations; plus V2 gate for corrected Eq (4.10))
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from scipy.linalg import eig, fractional_matrix_power as fmp, sqrtm
rng = np.random.default_rng(11)
def comm(A,B): return A@B - B@A
def C_of(K): return comm(K, K.conj().T)
def Phi(K):
    C = C_of(K); return 0.5*np.real(np.trace(C @ C.conj().T))

# ----- FIGURE 1: corrected relaxation flow -----
n = 6
K0 = rng.standard_normal((n,n)) + 1j*rng.standard_normal((n,n))
w0 = np.sort_complex(np.linalg.eigvals(K0)); S2 = np.sum(np.abs(w0)**2)
def kap(K):
    w, vl, vr = eig(K, left=True, right=True)
    i = np.lexsort((w.imag, w.real))
    return np.array([1.0/abs(vl[:,j]).conj()@vr[:,j]) for j in i[np.argsort(i)]], w
def f(M): return -2*comm(C_of(M), M)
h = 5e-5; Kt = K0.copy(); t = 0.0
ts, phis, hens, drifts, kmax = [], [], [], [], []
next_rec = 0.0
while t < 2.0:
    if t >= next_rec:
        w = np.sort_complex(np.linalg.eigvals(Kt))

```

```

        kv,_ = kap(Kt)
        ts.append(t); phis.append(Phi(Kt)); hens.append(np.linalg.norm(Kt)**2 - S2)
        drifts.append(np.max(np.abs(w - w0))); kmax.append(np.max(kv))
        next_rec = max(1e-3, next_rec*1.35) if next_rec > 0 else 1e-3
        k1=f(Kt); k2=f(Kt+0.5*h*k1); k3=f(Kt+0.5*h*k2); k4=f(Kt+h*k3)
        Kt = Kt + (h/6)*(k1+2*k2+2*k3+k4); t += h
fig, ax = plt.subplots(1, 2, figsize=(9, 3.4))
ax[0].semilogy(ts, phis, 'b-', label=r'$\Phi(K)$ (spectral stress)')
ax[0].semilogy(ts, hens, 'r--', label=r'$\Delta_F^2$ (Henrici budget)')
ax[0].semilogy(ts, np.maximum(drifts, 1e-16), 'g:', label='eigenvalue drift')
ax[0].set_xlabel('t'); ax[0].set_title('Isospectral relaxation: stress dies, spectrum frozen')
ax[0].legend(fontsize=8); ax[0].set_ylim(1e-13, 1e3)
ax[1].plot(ts, kmax, 'k-')
ax[1].axhline(1.0, color='gray', lw=0.7)
ax[1].set_xlabel('t'); ax[1].set_ylabel(r'$\max_i \kappa_i$')
ax[1].set_title(r'Co-vector/vector alignment: $\kappa_i$ to 1$')
plt.tight_layout(); plt.savefig('fig1_flow.pdf'); plt.close()
print("FIG1 done: final Phi", phis[-1] if phis else None, " final drift", drifts[-1])

# ----- FIGURE 2: area law census -----
def band_projector(H):
    ev, V = np.linalg.eigh(H); occ = ev < 0
    return V[:, occ] @ V[:, occ].conj().T
def census(P, idx, eps=1e-3):
    nu = np.linalg.eigvalsh(P[np.ix_(idx, idx)])
    return int(np.sum((nu > eps) & (nu < 1-eps)))
Ntot = 1200; th = 0.4
def Hid(m):
    H = np.zeros((Ntot, Ntot), complex); ph = np.exp(1j*th/Ntot)
    for j in range(Ntot):
        H[j,(j+1)%Ntot] = -ph; H[(j+1)%Ntot,j] = -np.conj(ph); H[j,j] = m*(-1)**j
    return H
Ls1 = [50,100,200,400]
N_gap = [census(band_projector(Hid(0.5)), np.arange(L)) for L in Ls1]
N_free = [census(band_projector(Hid(0.0)), np.arange(L)) for L in Ls1]
Lx = Ly = 44; Ns = Lx*Ly
def idx2(x,y): return (x%Lx)*Ly + (y%Ly)
def H2d(m):
    H = np.zeros((Ns, Ns), complex); px, py = np.exp(1j*0.37/Lx), np.exp(1j*0.83/Ly)
    for x in range(Lx):
        for y in range(Ly):
            i = idx2(x,y)
            H[i, idx2(x+1,y)] += -px; H[idx2(x+1,y), i] += -np.conj(px)
            H[i, idx2(x,y+1)] += -py; H[idx2(x,y+1), i] += -np.conj(py)
            H[i,i] = m*(-1)**(x+y)
    return H
Ls2 = [6,10,14,18,22]
P2g = band_projector(H2d(1.0)); P2f = band_projector(H2d(0.0))
def reg(L): return np.array([idx2(4+x,4+y) for x in range(L) for y in range(L)])
N2g = [census(P2g, reg(L)) for L in Ls2]
N2f = [census(P2f, reg(L)) for L in Ls2]
fig, ax = plt.subplots(1, 2, figsize=(9, 3.4))
ax[0].plot(Ls1, N_gap, 'bo-', label='gapped m=0.5')
ax[0].plot(Ls1, N_free, 'rs--', label='gapless m=0')
ax[0].set_xlabel('region length L'); ax[0].set_ylabel(r'$N_{\rm track}$')
ax[0].set_title('1D census: area law = constant'); ax[0].legend(fontsize=8); ax[0].set_ylim(0, 14)
ax[1].plot(Ls2, N2g, 'bo-', label='gapped m=1')
ax[1].plot(Ls2, N2f, 'rs--', label='gapless m=0')
ax[1].plot(Ls2, [4*L for L in Ls2], 'k:', label=r'$N=4L$ (unit density)')
ax[1].set_xlabel('region size L'); ax[1].set_title(r'2D census: $N$ to $perimeter$')
ax[1].legend(fontsize=8)
plt.tight_layout(); plt.savefig('fig2_arealaw.pdf'); plt.close()
print("FIG2 done:", N_gap, N_free, N2g, N2f)

# ----- FIGURE 3: refraction dynamics -----
eps_layer = 0.05
def traj(R1, R2, th1_deg, h=5e-5, tmax=6.0):
    Rf = lambda w: (R1+R2)/2 + (R2-R1)/2*np.tanh(w/eps_layer)

```

```

dRf = lambda w: (R2-R1)/2/np.cosh(w/eps_layer)**2/eps_layer
th1 = np.radians(th1_deg)
s = np.array([-1.0, np.sin(th1), np.cos(th1)]); t = 0.0
xs, ys = [0.0], [-1.0]
def g(s):
    w, vp, vq = s; R, dR = Rf(w), dRf(w)
    return np.array([vq, -dR*vq*vp/R, dR*(vp**2-vq**2)/(2*R)])
x = 0.0
while t < tmax:
    k1=g(s); k2=g(s+0.5*h*k1); k3=g(s+0.5*h*k2); k4=g(s+h*k3)
    s = s + (h/6)*(k1+2*k2+2*k3+k4); x += s[1]*h; t += h
    if len(xs) < 4000 and int(t/h) % 30 == 0: xs.append(x); ys.append(s[0])
    if s[0] > 1.0 or (s[0] < -1.0 and s[2] < 0): break
    return np.array(xs), np.array(ys)
fig, ax = plt.subplots(figsize=(6.4, 4))
for (R1,R2,th,lab,col) in [(1,4,50,'R 1->4, 50 deg (transmit)','b'),
                           (4,1,20,'R 4->1, 20 deg (transmit)','g'),
                           (4,1,45,'R 4->1, 45 deg (total reflection)','r')]:
    xs, ys = traj(R1,R2,th)
    ax.plot(xs, ys, col, label=lab)
ax.axhspan(-3*eps_layer, 3*eps_layer, color='gray', alpha=0.3)
ax.set_xlabel(r'$w_{\parallel}$'); ax.set_ylabel(r'$w_{\perp}$')
ax.set_title(r'EL dynamics through the rigidity layer: $R\sin^2\theta$ conserved to $10^{-13}$')
ax.legend(fontsize=8); plt.tight_layout(); plt.savefig('fig3_refract.pdf'); plt.close()
print("FIG3 done")

# ----- FIGURE 4: ledger diminishing returns -----
d = 5
Q = rng.standard_normal((d,d)); A0 = np.eye(d) + 0.05*(Q+Q.T)@(Q+Q.T).T/d
U, _ = np.linalg.qr(rng.standard_normal((d,d)))
def dR_aff(A,B):
    Ai = fmp(A,-0.5); ev = np.linalg.eigvalsh(Ai@B@Ai)
    return np.sqrt(np.sum(np.log(ev)**2))
def dS(A,B):
    s = np.linalg.slogdet((A+B)/2)[1] - 0.5*np.linalg.slogdet(A)[1] - 0.5*np.linalg.slogdet(B)[1]
    return np.sqrt(max(s,0))
lks, ratios = [1,2,4,6,8], []
for lk in lks:
    A1 = U @ np.diag(np.exp(np.linspace(-lk/2, lk/2, d))) @ U.T
    A0h = sqrtm(A0).real; A0hi = np.linalg.inv(A0h); Mid = A0hi@A1@A0hi
    npth = 300
    path = [A0h @ fmp(Mid, s).real @ A0h for s in np.linspace(0,1,npth+1)]
    accS = sum(dS(path[k], path[k+1]) for k in range(npth))
    ratios.append(accS/dS(A0, A1))
fig, ax = plt.subplots(figsize=(5.2, 3.4))
ax.plot(lks, ratios, 'mo-')
ax.axhline(1.0, color='gray', lw=0.7)
ax.set_xlabel(r'$\log \kappa$ (rigidity contrast)'); ax.set_ylabel('accumulated / direct')
ax.set_title('Ledger metric: diminishing returns grow with contrast')
plt.tight_layout(); plt.savefig('fig4_color.pdf'); plt.close()
print("FIG4 done:", [f"{r:.3f}" for r in ratios])

# ----- V2 GATE: corrected Eq (4.10) -----
l = np.zeros((8,3,3), complex)
l[0][0,1]=l[0][1,0]=1; l[1][0,1]=-1j; l[1][1,0]=1j; l[2][0,0]=1; l[2][1,1]=-1
l[3][0,2]=l[3][2,0]=1; l[4][0,2]=-1j; l[4][2,0]=1j; l[5][1,2]=l[5][2,1]=1
l[6][1,2]=-1j; l[6][2,1]=1j; l[7]=np.diag([1,1,-2])/np.sqrt(3)
fabc = np.zeros((8,8,8))
for a in range(8):
    for b in range(8):
        cab = comm(l[a],l[b])
        for c in range(8): fabc[a,b,c] = (np.trace(cab@l[c])/(4j)).real
psi = rng.standard_normal(8) + 1j*rng.standard_normal(8)
A = sum(psi[a]*l[a] for a in range(8))
direct = -2*comm(comm(A, A.conj().T), A)
formula = 8j*sum(fabc[a,b,c]*fabc[c,d,e]*np.imag(psi[a]*np.conj(psi[b]))*psi[d]*l[e]
                for b in range(8) for c in range(8) for d in range(8) for e in range(8))
print("V2-GATE corrected (4.10): ||direct - formula|| =", np.linalg.norm(direct - formula))

```


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